



M-Tech Production & Marketing

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TASK 1 REPORT

Internship Program — Batch 2026

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1. Introduction

This report presents the work completed for Task 1 (Experiment 07) of the M-Tech Production & Marketing AI/ML Internship Program, Batch 2026. This task covers Model Evaluation and Validation strategies, which are critical for building reliable AI systems. M-Tech Production & Marketing, based in Haripur, KPK, structures this lab to help us understand how to properly evaluate models, read classification errors, balance the bias-variance tradeoff, and use cross-validation techniques.

2. Objectives

- Learn how to correctly split data into training, validation, and testing sets.
- Understand how to calculate Precision, Recall, and F1-score from a Confusion Matrix.
- Learn the concepts of underfitting, overfitting, and the bias-variance tradeoff. Implement K-fold cross-validation to get stable model scores.
- Plot and interpret the ROC Curve and the Area Under the Curve (AUC).

3. Task Description

The task involves checking how well machine learning models generalize to new data. We need to split data into three separate partitions, measure errors using classification matrices, and show how models overfit when they become too complex. The task also requires running cross-validation loops and creating visual ROC charts to measure predictive performance accurately.

4. Work Done

I completed the lab experiments and implemented all 5 tasks within a Jupyter Notebook. I coded a 70/15/15 data split routine and verified mathematical calculations for standard metrics like F1-score. I generated a high-degree mathematical curve to visually demonstrate overfitting errors, built a Cross-Validation routine using Random Forests, and plotted a complete ROC curve showing the exact AUC percentage.

5. Challenges Faced

Precision vs. Recall: Remembering which metric handles False Positives versus False Negatives can be confusing. Precision focuses on keeping predictions clean, while Recall focuses on catching all targets.

Polynomial Overfitting Setup: Simulating a clear mathematical overfit using high-degree equations requires carefully generating noisy data blocks so the error gap shows up correctly.

6. Key Learnings

The Test Set Rule: The test set must stay completely unseen. If the model looks at it during training or tuning, the final evaluation becomes biased and unreliable.

Overfitting vs. Underfitting: Overfitting happens when a model memorizes data noise (high variance). Underfitting happens when a model is too simple to learn the pattern (high bias).

Cross-Validation: Instead of relying on one lucky data split, Cross-Validation splits the data multiple times (K-times) and averages the scores, giving a much more honest performance check.

7. Conclusion

This lab successfully highlighted that training accuracy is not the final goal in machine learning. By utilizing tools like confusion matrices, cross-validation, and ROC-AUC curves, I learned how to identify if a model is simply memorizing examples or actually learning useful rules. These evaluation practices ensure that AI models remain accurate and dependable when deployed for real-world users.

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